

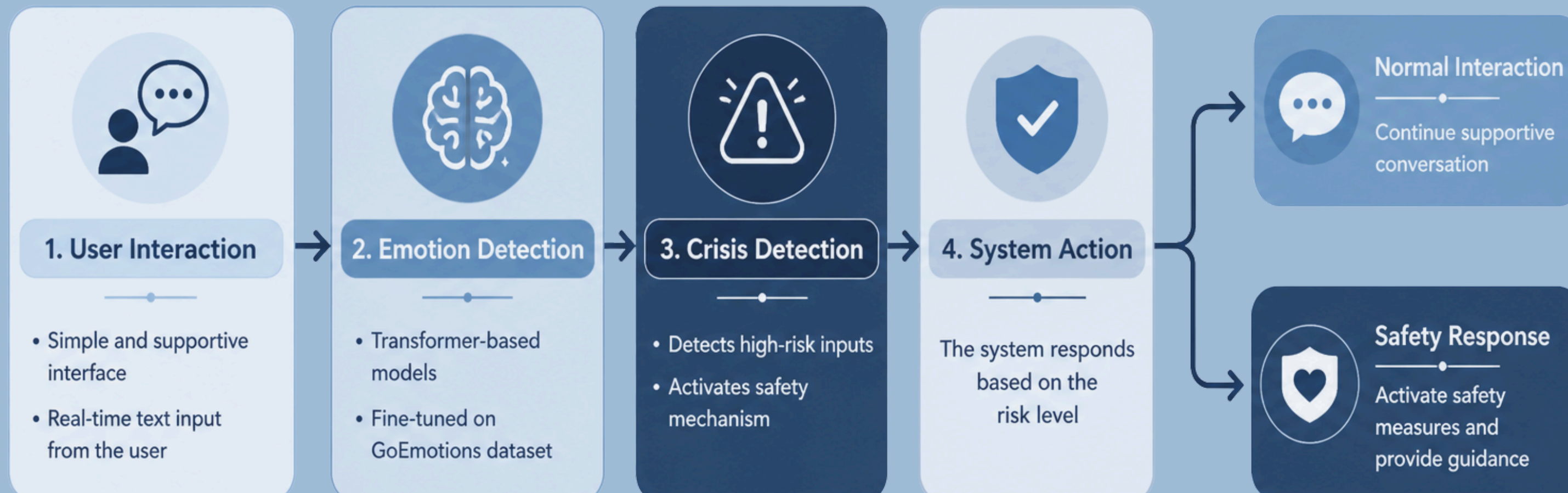


INTRODUCTION

More than **one billion** people worldwide suffer from mental health conditions, yet **75%** receive no treatment due to social stigma, high costs, and a shortage of specialists.

EmpathyBot is an AI-powered companion that bridges this gap through real-time emotion detection and crisis management — making mental health support accessible to everyone, everywhere .

METHODOLOGY



Workflow of EmpathyBot illustrating the flow from user input to system analysis.

EXPERIMENTS

Dataset & Preprocessing

- Dataset: GoEmotions (28 emotion categories, multi-label)
- Split: 80% train / 10% val / 10% test (43,410 / 5,426 / 5,427 samples)
- Tokenization: RoBERTa BPE tokenizer, max length = 128 tokens
- Labels: Multi-hot encoding for 28 emotion classes

Training Strategy

- Loss function: Weighted Focal Loss ($\gamma = 2.0$)
- Optimizer: AdamW (lr: 1e-5 backbone, 5e-5 head)
- Layer-wise Learning rate Decay (LLRD): decay factor 0.95
- Batch size: 16
- Epochs: 8 (early stopping at epoch 4)

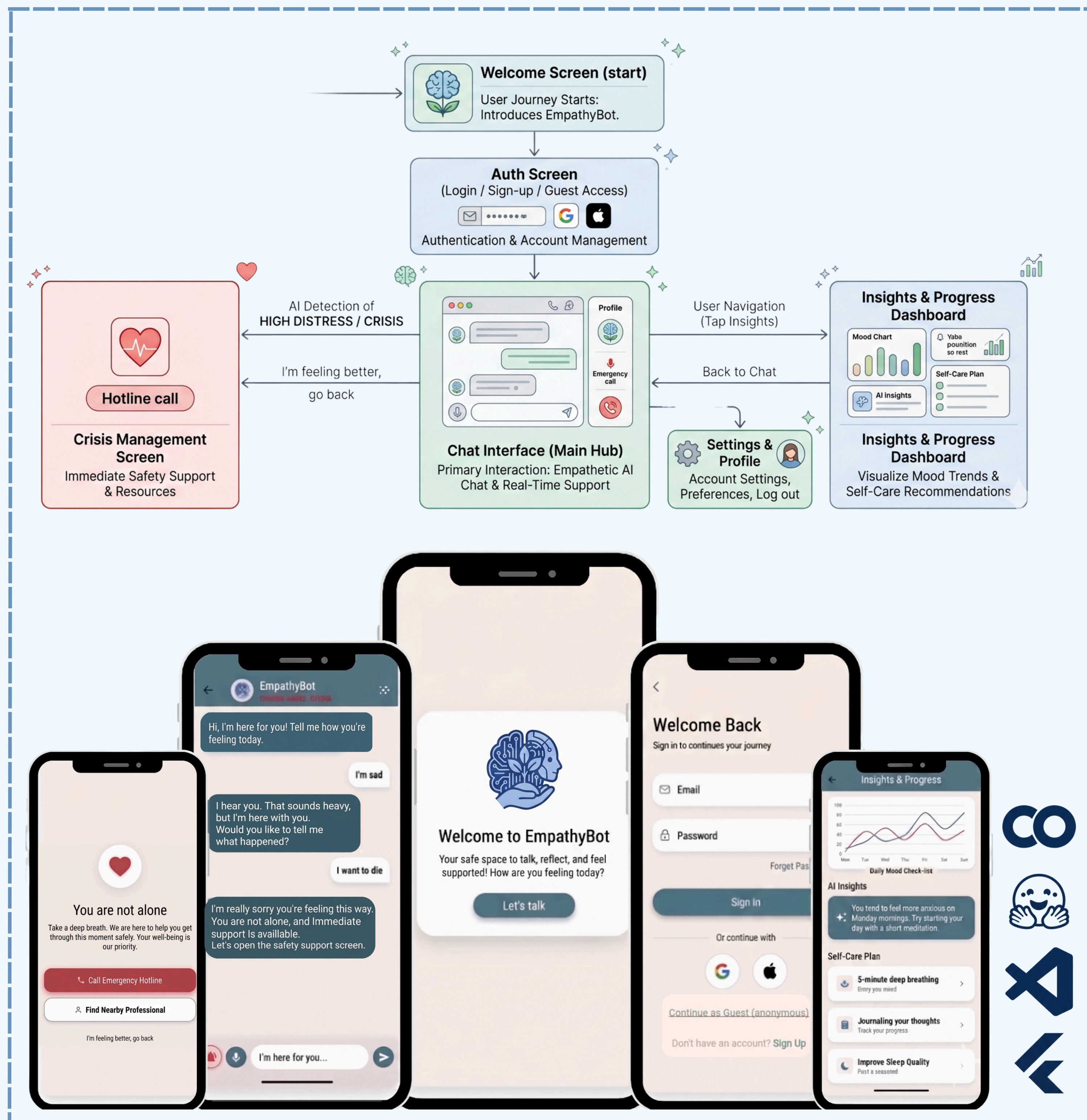
Model Architecture (Final)

- Backbone: RoBERTa-Large (355M parameters)
- Classification head: Linear layer + Multi-Sample Dropout (5 masks, rates 0.1–0.5)
- Output activation: Sigmoid (multi-label)

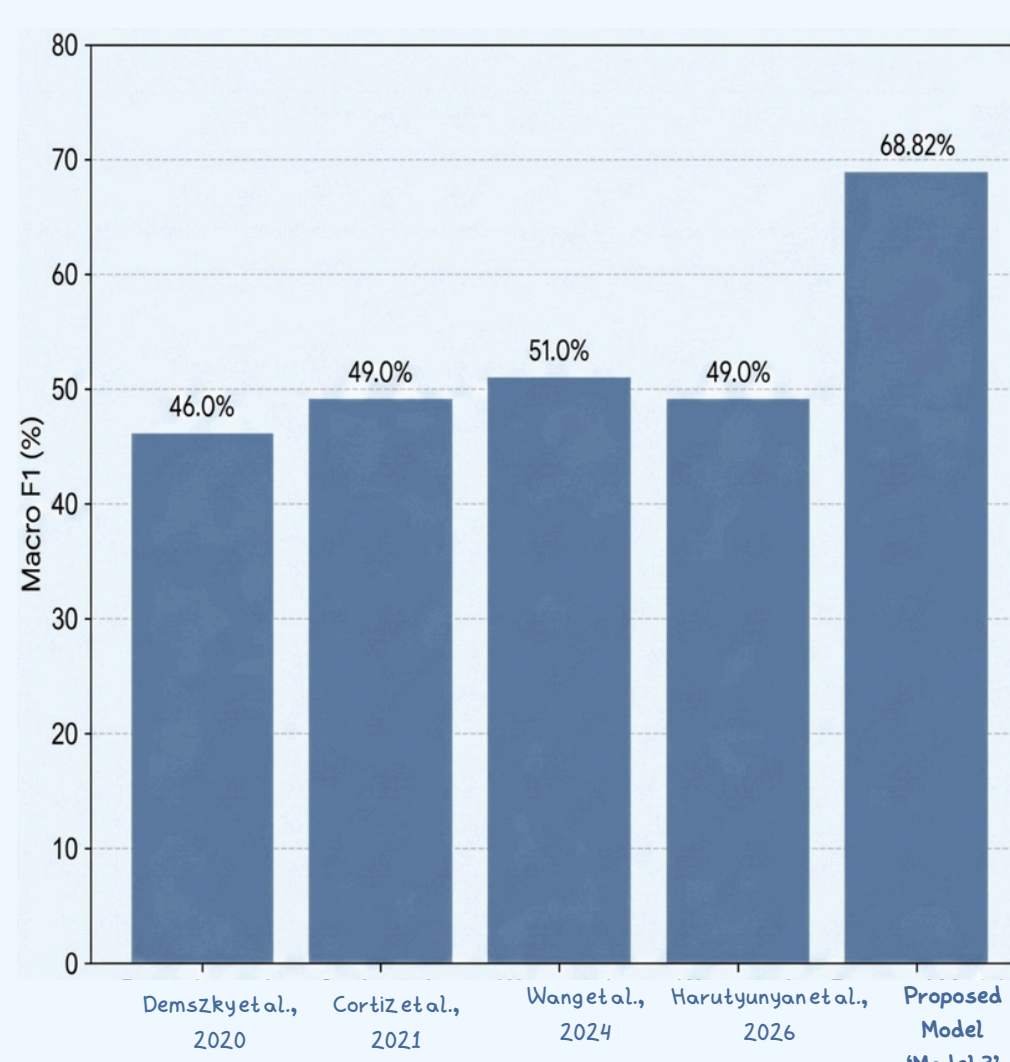
Comprehensive Evaluation of Baseline and Proposed Model Iterations

Aspect	Baseline	Model 1	Model 2 (v1)	Model 2 (v2)	Model 3 (Final)
Base Architecture	RoBERTa-Large	RoBERTa-Large	RoBERTa-Large	RoBERTa-Large	RoBERTa-Large
Data Split (Train/Val/Test)	80/10/10	70/20/10	70/20/10	80/10/10	80/10/10
Optimization Strategy	Cross Entropy	Focal Loss ($\gamma = 2$)	WFL + MSD	WFL + MSD	Hybrid Opt.
Macro F1 (Validation)	47.55%	53.85%	54.53%	50.49%	54.84%
Macro F1 (Testing)	48.81%	59.73%	58.21%	54.65%	68.82%

See More about our proposed model



Comparison of Macro-F1 'Testing' Scores across Different Studies



CONCLUSION

EmpathyBot successfully demonstrates that AI can bridge the mental health treatment gap by providing immediate and accessible emotional support. Through real-time emotion detection and crisis management, the system proves both effective and ethical. Future work includes expanding multilingual capabilities.